

Ziang Li

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Education

Oct 2025 - Present	University of Regensburg Master's studies in mathematics; Regensburg, Germany Master thesis in progress: An Isomorphism Between the Blow-Up of the Deligne–Mumford Moduli Space and the Decorated Deligne–Mumford Moduli Space. Supervisor(s): Prof. Stefan Friedl and Prof. Bernd Ammann Selected coursework: <i>Grades in Germany are on a scale from 1.0 to 5.0; 5.0 is a failing grade.</i> Twisted homology and Reidemeister torsion (1.3) Simple homotopy theory and manifold topology (1.0) Seminar on Floer homology, Part II (1.0)	Algebraic Topology I (1.3) Algebraic Topology II (1.0) Characteristic classes (1.3)
	Xi'an Jiaotong University Bachelor's degree in mathematics; Xi'an, China Thesis: Quasi-isometry Groups of $\mathbb{R}^n$. Supervisor: Qiang Zhang. Selected coursework: Riemannian Geometry (100) Graph Theory and Combinatorics (92) Mathematical Analysis Seminar (99)	Functional Analysis (90) Algebra Seminar (92) Differentiable Manifolds (90)

Research interests

Geometric topology, especially knot theory and the topology of 3- and 4-manifolds.

Academic experience

Seminar talks

Jul 2026	Proof of the s-cobordism theorem Simple homotopy theory and manifold topology seminar, University of Regensburg Organizer: Dr. Marco Volpe. Report
Jul 2026	A surgery formula for Seiberg-Witten invariants Seminar on spin geometry, University of Regensburg Organizer: Prof. Bernd Ammann. Report
Dec 2025	Proof of the Arnold conjecture Seminar on Floer homology, Part II, University of Regensburg Organizer: Prof. Bernd Ammann. Report
Nov 2025	An introduction to Seiberg-Witten theory and its applications to 4-manifolds Surfaces in 4-manifolds seminar, University of Regensburg Organizer: Prof. Stefan Friedl. Report

Research notes

Jul 2026	Direct proof of equivalence of two definitions of Alexander polynomials Supervised by: Prof. Stefan Friedl. I gave a direct proof that the definitions of the Alexander polynomial via the Burau representation and Fox calculus are equivalent.
Sep 2026	Master thesis in progress: An Isomorphism Between the Blow-Up of the Deligne–Mumford Moduli Space and the Decorated Deligne–Mumford Moduli Space. Supervised by: Prof. Bernd Ammann. Tba

Conferences, summer schools, and workshops

8 Jun 2026 - 12 Jun 2026	Masterclass: Seiberg-Witten Invariants University of Copenhagen; Copenhagen, Denmark. Program
18 May 2026 - 22 May 2026	Spring school "Khovanov and Floer homotopy" Nantes University; Nantes, France. Program
27 Dec 2025 - 28 Dec 2025	ALGANT Alumni in China University of Science and Technology of China; Hefei, China. Program
8 Dec 2025 - 12 Dec 2025	Workshop on surfaces in 4-manifolds University of Regensburg; Regensburg, Germany. Program
1 Jul 2025 - 11 Jul 2025	USTC's 2025 Summer School in Geometry and Analysis University of Science and Technology of China; Hefei, China. Program
22 Aug 2024 - 28 Aug 2024	SUSTech's 2024 Number Theory and Arithmetic Geometry Conference Southern University of Science and Technology; Shenzhen, China. Program
29 Jul 2024 - 9 Aug 2024	BICMR's 2024 Summer School in Differential Geometry Beijing International Center for Mathematical Research, Peking University; Beijing, China. Program
1 Jul 2024 - 9 Jul 2024	YMSC's 2024 Summer School on Frontiers in Topology Yau Mathematical Sciences Center, Tsinghua University; Beijing, China. Program

Student-led reading seminars

Sep 2024 - Jun 2025	Symplectic geometry (Organizer and main speaker)
Mar 2024 - Jun 2024	Characteristic classes (Organizer and main speaker)
Dec 2023 - Mar 2024	Morse theory (Organizer and main speaker)
Sep 2023 - Jun 2024	Category theory and homological algebra (Participant)
Mar 2022 - Jan 2023	Algebraic topology (Participant and speaker)
Feb 2022 - Jun 2022	Riemannian geometry (Organizer and main speaker)
Sep 2021 - Jan 2022	Differential manifolds and de Rham cohomology (Organizer and main speaker)

References

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